

EXPERIMENT-2

ROLL NO: 240701058

NAME: ASHWIN C

USER INTERFACE AND DESIGN

Design a UI where users recall visual elements (e.g., icons).
Evaluate the effect of chunking on user memory.

FRAME 1:

INSTRUCTION FRAME:

Chunking Analysis of the Instruction Page :

Chunking is a cognitive strategy that breaks down information into smaller, manageable units, making it easier to process and retain. The Memory Recall Task instruction page effectively utilizes chunking in the following ways:

1. Clear and Sequential Numbering :

- The instructions are broken down into six steps, making it easier to follow.
- Each step presents one key action in a structured manner, reducing cognitive overload.

2. Logical Grouping of Information :

- Observation Phase (Steps 1-2) -> Users learn about what they will see.
- Memorization Strategy (Step 3) -> Encourages users to focus on remembering items.
- Recall Phase (Steps 4-5) -> Explains how users will recall information.

3. Visual Hierarchy and Design Elements :

- Bold, large title ("MEMORY RECALL TASK") -> Grabs attention and clearly states the task.
- Bullet points and spacing -> Reduce clutter, enhancing readability.
- Highlighted "START" button -> Signals the next step, keeping navigation intuitive.
- Honey bee and honeycomb-themed visuals -> Add a playful touch, engaging users without distraction.

4. Time Constraint Reinforcement :

- Step 5 explicitly states, "You have 5 seconds to view the items."
- This reinforces expectations while subtly urging users to focus.
- A Progress bar also decreases for each second ensuring user to focus on time constraint.

5. Simplicity and Clarity :

- Instructions use short, direct sentences, avoiding unnecessary complexity.
- The active voice makes it more engaging and action-oriented.
- More clarity helps users to remember the icons.



Figure 1: Instruction Frame - Memory Recall Task

FRAME 2:

CHUNKING PHASE:

Analysis of the Memory Recall Task - Chunking Phase Screen :

This screen represents the Chunking Phase of a Memory Recall Task, where users observe and memorize different icons within a limited time. Below is a breakdown of its key components:

1. Purpose of the Screen

- This is the visual memory encoding phase, where users view and group items mentally before recalling them.
- The term "Chunking Phase" suggests that the game applies the chunking principle, which helps users remember items by categorizing them into meaningful groups.

2. Key Elements and UI Components :

- **Countdown Timer (Top Left - Pink Circle: "00:05")** :Indicates that users have 5 seconds to observe and memorize the displayed items.
- **The bright pink colour and bold text create urgency, ensuring users stay focused.**
- **Progress Bar (White Bar Below Title)** :Shows the time remaining visually, reinforcing the sense of urgency.
- **A partially filled bar suggests that some time has already passed.**
- **Grid of Icons** :Various icons (such as a pig, burger, bee, snowman, police car, arrow, laughing emoji, etc.) are displayed in a 5x4 grid format.

- These icons are visually distinct yet grouped by similarities, encouraging chunking strategies.
- Categorizing by theme (e.g., animals, food, objects).
- Grouping similar colours or backgrounds (e.g., pink and blue tiles).
- Associating repeated icons (e.g., multiple bees, burgers, pigs).

3. How the Chunking Phase Works :

- Users scan the grid and look for patterns or related items to create mental chunks.
- The countdown timer limits observation time, forcing quick memory strategies.
- Once time is up, users transition to the recall phase, where they must identify previously seen items.

4. Cognitive and UX Benefits of Chunking :

- Enhances short-term memory by allowing users to recall groups of information instead of individual elements.
- Reduces cognitive overload by helping users organize data efficiently.
- Improves pattern recognition, making recall easier and more accurate.



Figure 2: Chunking Phase Screen

FRAME 3:

RECALL PHASE:

Analysis of the Memory Recall Task - Selection Phase :

This screen represents the Selection Phase of a Memory Recall Task, where users recall and choose the items they remember from the previous Chunking Phase. Below is a breakdown of its components:

1. Purpose of the Screen :

- This is the memory retrieval stage, where users select the items they remember seeing in the previous phase.
- The goal is to test the effectiveness of chunking and short-term memory retention.

2. Key Elements and UI Components :

- **Title & Instructions** : "MEMORY RECALL TASK" (Bold Header) - Reinforces the purpose of the task.
- **"SELECT THE ITEMS YOU REMEMBER:"** - Clear instruction guiding the user to choose remembered items.
- **Grid of Icon Choices** : A set of eight icons are presented as multiple-choice options.
- **Some icons were previously displayed, while others are distractors (new icons added to confuse users).**
- **Users must identify which icons appeared in the Chunking Phase.**
- **Check Box** : A component set containing checkboxes of empty and filled checkboxes are created.
- **The component set is renamed as TRUE and FALSE respectively.**
- **The asset is added in frame.**
- **"SUBMIT" Button (Bottom Center)** : Once users have made their selections, they press "SUBMIT" to confirm their recall choices.

3. How the Selection Phase Works :

- Users analyze the displayed icons and recall which ones they saw in the Chunking Phase.
- They select the remembered icons using the radio buttons below each option.
- Some icons are distractors, testing whether the user's memory is accurate or if they mistakenly recall incorrect icons.
- Clicking "SUBMIT" finalizes their choices, leading to a results or feedback screen.

4. Cognitive and UX Benefits :

- Tests memory accuracy by comparing user selections with previously displayed items.
- Incorporates distractors to evaluate how well users distinguish real vs. false memories.



Figure 3: Recall Phase - Selection Screen

FRAME 4:

REMEMBER PHASE:

Analysis of the Memory Recall Task - Score & Feedback Screen :

This screen represents the Score & Feedback Phase of the Memory Recall Task, where users receive their performance evaluation based on the selections made in the previous Recall Phase.

1. Purpose of the Screen :

- Provides feedback on recall accuracy by showing the number of correct answers.
- Allows users to decide their next action (continue, restart, or exit).

2. Key Elements & UI Components :

- **Title & Score Display** : "MEMORY RECALL TASK" - Reinforces the game's title.
- **"YOUR RECALL SCORE!"** - Indicates that the user's performance is being displayed.
- **Action Buttons (Right Side)** : "RESTART" - Allows the user to retake the memory recall test.
- **"EXIT"** - End Task and Exits Game.

3. How This Phase Works :

- A variable (number) is created and assigned to score value.
- Another prototype is added to checkbox present in "RECALL FRAME".
- The prototype action is increased or decreased as per image in both frames.
- It calculates the accuracy score and displays it on a digital number.
- Users review their performance and choose their next action either Continue or Exit.

4. Cognitive & UX Benefits :

- Instant feedback helps users track their memory performance.
- Multiple options (Continue, Exit) give users control over their learning experience.



Figure 4: Score & Feedback Screen

PROTOTYPE LINK:

[chunking – Figma](#)